

ON A CONSTANT RANK THEOREM FOR SPECIAL LAGRANGIAN TYPE OPERATOR ON $\mathbb{S}^n$

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ABSTRACT. In this note, we study the full rank property of a class of in-homogeneous fully nonlinear elliptic equations of special Lagrangian type on $\mathbb{S}^n$. Our approach is based on the application of the Lagrange multiplier method, which allows us to derive the necessary differential inequality. We then employ the strong maximum principle together with the Minkowski integral formula to establish the result. As an application, combining the existence result of Guan and Zhang [10], we obtain the existence of convex solutions to the Christoffel-Minkowski type problem for special Lagrangian type equations on $\mathbb{S}^2$ under a structural condition.

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1. INTRODUCTION

The celebrated work of Guan and Ma [8] solved the classical Christoffel-Minkowski problem in differential geometry, which concerns finding convex solutions to the k -Hessian equation

$$(1.1) \quad \sigma_k(W) = f(x) \quad \text{on } \mathbb{S}^n,$$

for $k = 1, 2, \dots, n$, where $W = (w_{ij}) = (u_{ij} + u\delta_{ij})$ is the spherical Hessian associated with a function $u : \mathbb{S}^n \rightarrow \mathbb{R}$. Here, u_{ij} denotes the second order covariant derivatives of u with respect to a local orthonormal frame $\{e_1, e_2, \dots, e_n\}$ on $\mathbb{S}^n$, and $f(x)$ is a smooth function satisfying the condition $\int_{\mathbb{S}^n} x_i f(x) dx = 0$ for all $i = 1, 2, \dots, n+1$. The function σ_k denotes the k -th elementary symmetric function, the precise definition of which will be recalled in Section 2.

A function u is said to be convex solution of (1.1) if the associated matrix W is positive definite on $\mathbb{S}^n$. Accordingly, a key step in establishing the existence of convex solutions in [8] was the proof of a constant rank theorem for equation (1.1), which asserts that W is of full rank on $\mathbb{S}^n$ provided that the matrix $((f^{-1/k})_{ij} + f^{-1/k}\delta_{ij})$ is semi-positive definite. The constant rank theorem was first introduced and developed in the 1980s by Caffarelli and Friedman [5], Singer, Wong, Yau and Yau [19], and Korevaar and Lewis [14] during their investigations into the convexity of solutions to semilinear elliptic equations.

After the contributions of Guan and Ma [8], a number of authors have further developed constant rank theorem in the context of Christoffel-Minkowski type problems; see, for example [12, 9, 4, 3, 16, 13, 6, 7], and the references therein. In [10], Guan and Zhang considered a more general fully nonlinear in-homogeneous elliptic equation on $\mathbb{S}^n$ and proved the existence of $\mathcal{G}$ -invariant admissible solutions

$$(1.2) \quad \sigma_k(W) + \alpha \sigma_{k-1}(W) = \sum_{l=0}^{k-2} \alpha_l \sigma_l(W) \quad \text{on } \mathbb{S}^n,$$

in which α and $\alpha_l > 0$ with $l = 0, 1, \dots, k-2$ are given smooth $\mathcal{G}$ -invariant functions on the sphere. Here, $\mathcal{G}$ -invariance means that there exists an automorphic group $\mathcal{G}$ of $\mathbb{S}^n$ without fixed points such that $\alpha(g(x)) = \alpha(x)$ and $\alpha_l(g(x)) = \alpha_l(x)$ for all $g \in \mathcal{G}$ and $x \in \mathbb{S}^n$. A solution of (1.2) is called admissible if the eigenvalue $\lambda(W) = (\lambda_1, \lambda_2, \dots, \lambda_n)$ lie in the cone

$$\Gamma_{k-1} = \{\lambda \in \mathbb{R}^n \mid \sigma_1(\lambda) > 0, \sigma_2(\lambda) > 0, \dots, \sigma_{k-1}(\lambda) > 0\}.$$

However, due to the lack of a constant rank theorem for (1.2), the existence of $\mathcal{G}$ -invariant convex solutions (i.e., $\lambda(W) \in \Gamma_n$) remained unknown.

As pointed out in [10], establishing the constant rank theorem for in-homogeneous fully nonlinear equation on $\mathbb{S}^n$ is a highly interesting and challenging problem. Motivated by this, in this note we study the constant rank result for a class of in-homogeneous Christoffel-Minkowski type equation of the form

$$(1.3) \quad \sum_{i=1}^n \arctan \lambda_i(W(x)) = \Theta(x) \quad \text{on } \mathbb{S}^n$$

under suitable structural conditions, where $\lambda_i(W)$ are the eigenvalues of W and $\Theta \in (0, n\pi/2)$ is a given function on the sphere, which we call the phase. This equation can be viewed as a natural extension of the special Lagrangian equation originally introduced by Harvey and Lawson [11]. For this reason, we sometimes refer to the operator

$$(1.4) \quad F(W) \triangleq \sum_{i=1}^n \arctan \lambda_i(W),$$

where W is a symmetric matrix, as being of special Lagrangian type.

It is worth mentioning that Bryan, Ivaki and Scheuer [3] proved a constant rank theorem for in-homogeneous fully nonlinear elliptic equations on Riemannian manifolds under certain structural conditions, using the viscosity approach. More recently, Chen and Xu [6] also studied the constant rank property of equation (1.3) for $\Theta \in (0, \pi/2 - \delta]$, where $\delta \in (0, \pi/2)$. We note, however, that the structural conditions in our work differ from theirs. Our proof is based on a careful application of the method of Lagrange multipliers. We now state the main result of this note.

Theorem 1.1. *Let $\Theta : \mathbb{S}^n \rightarrow (0, \pi/2)$ be a C^2 function. Assume that u is a C^4 solution of (1.3), with W is non-negative definite. If the matrix $((1/\tan \Theta)_{ij} + (1/\tan \Theta)\delta_{ij})$ is semi-positive definite for every $x \in \mathbb{S}^n$, then W is positive definite on $\mathbb{S}^n$.*

We observe that equation (1.3) can be equivalently reformulated as a combination of elementary symmetric functions, given explicitly by

$$(1.5) \quad \sin \Theta \sum_{k=0}^{[n/2]} (-1)^k \sigma_{2k}(W) + \cos \Theta \sum_{k=0}^{[(n-1)/2]} (-1)^k \sigma_{2k+1}(W) = 0.$$

In the case $n = 2$, it is easy to see that equation (1.5) corresponds to the special case of (1.2) with $k = 2$ and $n = 2$. Thus, according to the result of Guan and Zhang [10], the existence of $\mathcal{G}$ -invariant admissible solutions to equation (1.3) is known when $n = 2$, provided that $1/\tan \Theta$ is $\mathcal{G}$ -invariant. Combining this with Theorem 1.1, we derive the following corollary.

Corollary 1.2. *Let $\mathcal{G}$ be an automorphic group of $\mathbb{S}^2$ without fixed points. Assume that $1/\tan \Theta$ is a smooth $\mathcal{G}$ -invariant function on $\mathbb{S}^2$, where $\Theta \in (0, \pi/2)$, and that the spherical Hessian $((1/\tan \Theta)_{ij} + (1/\tan \Theta)\delta_{ij})$ is semi-positive definite for every $x \in \mathbb{S}^2$. Then equation (1.3) admits a $\mathcal{G}$ -invariant smooth, convex solution u when $n = 2$.*

Geometrically, for such Θ , there exists a strictly convex, smooth hypersurface $M \subset \mathbb{R}^3$ whose special Lagrangian curvature satisfies

$$(1.6) \quad \arctan \kappa_1 + \arctan \kappa_2 = \pi - \Theta$$

along its outer unit normal, where κ_1, κ_2 denote the principal curvatures of M .

We remark that equation (1.6) was first studied by Smith [20], who referred to it as the special Lagrangian curvature equation.

The authors find that the method of Lagrange multipliers can be used to establish a constant rank theorem for convex solutions of a Lagrangian phase equation

$$\sum_{i=1}^n \arctan \lambda_i(D^2u) = \Theta(x) \quad \text{on } \Omega \subset \mathbb{R}^n$$

under the structural condition that $1/\tan \Theta$ is convex, where $\lambda_i(D^2u)$ denote the eigenvalues of the Hessian D^2u , and $\Theta(x)$ is a smooth function on Ω taking values in $(0, \pi/2)$. The proof is the same as that of Theorem 1.1 and will therefore be omitted. This result shows that, when $\Theta \in (0, \pi/2)$, the convexity of $1/\tan \Theta$ is a weaker structural condition than the concavity of Θ required in earlier works such as Caffarelli, Guan and Ma [4]. In other words, while the concavity of Θ implies that $1/\tan \Theta$ is convex, the converse does not hold. More results on constant rank theorems for (convex) solutions of special Lagrangian type equations, or more generally fully nonlinear elliptic equations, have been studied extensively by many authors; see [1, 17, 4, 2, 21, 22, 15].

The organization of this note is as follows. In Section 2, we provide some preliminary results, including elementary symmetric functions, computations related to the special Lagrangian operator and so on. In Section 3, we establish a differential inequality (Lemma 3.1), which combined with the strong maximum principle and the Minkowski integral formula, leads to the proof of Theorem 1.1.

2. PRELIMINARIES

We recall the definition of k -th elementary symmetric functions. For $1 \leq k \leq n$ and $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_n) \in \mathbb{R}^n$, define

$$\sigma_k(\lambda) = \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} \lambda_{i_1} \lambda_{i_2} \dots \lambda_{i_k}.$$

We also make the convention that $\sigma_0(\lambda) = 1$ and $\sigma_{-1}(\lambda) = \sigma_{n+1}(\lambda) = 0$. The definition of σ_k for vectors in $\mathbb{R}^n$ can be extended to $n \times n$ symmetric matrices in a natural way by letting $\sigma_k(W) = \sigma_k(\lambda(W))$, where $\lambda(W) = (\lambda_1(W), \lambda_2(W), \dots, \lambda_n(W))$ are the eigenvalues of the symmetric matrix W . We denote by $\sigma_{k-1}(W|i)$ the $(k-1)$ -th elementary symmetric function with $\lambda_i = 0$ and $\sigma_{k-2}(W|ij)$ ($i \neq j$) the $(k-2)$ -th elementary symmetric function with $\lambda_i = \lambda_j = 0$. The following lemma provides the first and second order derivatives of the elementary symmetric functions.

Lemma 2.1 (Proposition 2.2 in [8]). *Suppose that $W = (w_{\alpha\beta})$ is diagonal. For $k = 1, 2, \dots, n$, then we have*

$$(2.1) \quad \frac{\partial \sigma_k(W)}{\partial w_{\alpha\beta}} = \begin{cases} \sigma_{k-1}(W|\alpha) & \text{if } \alpha = \beta, \\ 0 & \text{if } \alpha \neq \beta, \end{cases}$$

and

$$\frac{\partial^2 \sigma_k(W)}{\partial w_{\alpha\beta} \partial w_{\gamma\delta}} = \begin{cases} \sigma_{k-2}(W|\alpha\gamma) & \text{if } \alpha = \beta, \gamma = \delta, \alpha \neq \gamma, \\ -\sigma_{k-2}(W|\alpha\gamma) & \text{if } \alpha = \delta, \beta = \gamma, \alpha \neq \gamma, \\ 0 & \text{otherwise.} \end{cases}$$

Let F denote the special Lagrangian type operator given by (1.4). The first and second order derivatives of F are explicitly computable, as stated in the follows.

Lemma 2.2. *Suppose that W is diagonal, we have*

$$F^{\alpha\beta} = \frac{\partial F(W)}{\partial w_{\alpha\beta}} = \begin{cases} \frac{1}{1+\lambda_\alpha^2} & \text{if } \alpha = \beta, \\ 0 & \text{if } \alpha \neq \beta, \end{cases}$$

and

$$F^{\alpha\beta, \gamma\delta} = \frac{\partial^2 F(W)}{\partial w_{\alpha\beta} \partial w_{\gamma\delta}} = \begin{cases} -\frac{2\lambda_\alpha}{(1+\lambda_\alpha^2)^2} & \text{if } \alpha = \beta = \gamma = \delta, \\ -\frac{\lambda_\alpha + \lambda_\beta}{(1+\lambda_\alpha^2)(1+\lambda_\beta^2)} & \text{if } \alpha = \delta, \beta = \gamma, \\ 0 & \text{otherwise.} \end{cases}$$

We now turn to a fundamental result from linear algebra, whose proof is included for completeness.

Lemma 2.3. *Let $A = \text{diag}(a_1, a_2, \dots, a_n)$ be an invertible diagonal $n \times n$ matrix, and let $v = (v_1, v_2, \dots, v_n)^\top \in \mathbb{R}^n$. For any $c \in \mathbb{R}$, it holds that*

$$\det(A + cvv^\top) = \left(\prod_{i=1}^n a_i \right) \left(1 + c \sum_{i=1}^n \frac{v_i^2}{a_i} \right).$$

Proof. Since A is invertible, one can write

$$\det(A + cvv^\top) = \det(A) \cdot \det(I + cA^{-1}vv^\top).$$

It is therefore sufficient to compute the determinant of $I + cA^{-1}vv^\top$. Notice that the matrix $cA^{-1}vv^\top$ has rank one and therefore has exactly one non-zero eigenvalue. Since it is symmetric, the non-zero eigenvalue equals its trace and is given by

$$\lambda = \text{tr}(cA^{-1}vv^\top) = cv^\top A^{-1}v = c \sum_{i=1}^n \frac{v_i^2}{a_i}.$$

It follows that the eigenvalues of $I + cA^{-1}vv^\top$ consist of $1 + c \sum_{i=1}^n \frac{v_i^2}{a_i}$ and 1 with multiplicity $n - 1$. Hence, we get the desired formula. $\square$

3. PROOF OF THEOREM 1.1

We devote this section to the proof of Theorem 1.1. The strategy is to first derive a differential inequality in a neighborhood of the point where the spherical Hessian matrix $W = (w_{ij})$ attains its minimal rank, as stated in Lemma 3.1. This estimate (i.e., (3.1)) allows us to invoke the strong maximum principle, which leads to the conclusion that W has constant rank on $\mathbb{S}^n$. The full rank property then follows from the Minkowski integral formula.

Assume that the spherical Hessian W attains its minimal rank l at some point $x_0 \in \mathbb{S}^n$. We then select an open neighborhood $\mathcal{O}$ of x_0 and define

$$\phi(x) = \sigma_{l+1}(w_{ij}(x)) \quad \text{in } \mathcal{O}.$$

Lemma 3.1. *Let $\Omega \subset \mathbb{S}^n$ be an open subset, and let $\Theta \in C^2(\Omega)$ take values in $(0, \pi/2)$. Assume that $u \in C^4(\Omega)$ is a solution to (1.3) and that the spherical Hessian W is non-negative definite. Suppose that W attains minimal rank l at some point $x_0 \in \Omega$, let $\mathcal{O}$ be a neighborhood of x_0 . If the matrix $((1/\tan \Theta)_{ij} + (1/\tan \Theta)\delta_{ij})$ is semi-positive definite for all $x \in \mathcal{O}$, then there exists a constant $C > 0$, depending only on u , Θ , and n , such that*

$$(3.1) \quad \sum_{\alpha, \beta} F^{\alpha\beta} \phi_{\alpha\beta} \leq C(|\nabla \phi| + \phi) \quad \text{in } \mathcal{O}.$$

Proof. For any $x \in \Omega$, let $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ be the eigenvalues of W at x . There is a positive constant $c > 0$ such that $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_l \geq c$. Let $G = \{1, 2, \dots, l\}$ and $B = \{l+1, l+2, \dots, n\}$ be the “good” and “bad” sets of indices, respectively. If there is no confusion, we also denote the “good” and “bad” eigenvalues of W by G and B , respectively.

Following the notations in [5], for two functions $h(x)$ and $k(x)$ defined in $\mathcal{O}$, we say that $h(x) \lesssim k(x)$ if there exists a positive constant C_1 such that

$$h(x) - k(x) \leq C_1(|\nabla \phi(x)| + \phi(x)).$$

We also write $h(x) \sim k(x)$ if $h(x) \lesssim k(x)$ and $k(x) \lesssim h(x)$. Next, we write $h \lesssim k$ if the above inequality holds for any $x \in \mathcal{O}$, with constant C_1 independent of x . Finally, $h \sim k$ means $h \lesssim k$ and $k \lesssim h$.

For each $x \in \mathcal{O}$ fixed, by choosing a local orthonormal frame $\{e_1, e_2, \dots, e_n\}$, we suppose that W is diagonal at x and $w_{ii} = \lambda_i$ for any $i = 1, 2, \dots, n$. The following calculations will be carried out at a fixed point x . When we use the relation $\lesssim$, all the constants are under controlled.

Since W is diagonal, we have that

$$0 \sim \phi \sim \sigma_l(G) \sum_{i \in B} w_{ii} \sim \sum_{i \in B} w_{ii},$$

and

$$(3.2) \quad w_{ii} \sim 0, \quad i \in B.$$

It follows from (3.2) that

$$(3.3) \quad \sigma_l(\lambda|i) \sim \begin{cases} 0 & \text{if } i \in G, \\ \sigma_l(G) & \text{if } i \in B \end{cases}$$

holds, and moreover we get

$$(3.4) \quad \sigma_{l-1}(\lambda|ij) \sim \begin{cases} 0 & \text{if } i \in G, j \in G, \\ \sigma_{l-1}(G|i) & \text{if } i \in G, j \in B, \\ \sigma_{l-1}(G|j) & \text{if } i \in B, j \in G, \\ \sigma_{l-1}(G) & \text{if } i \in B, j \in B. \end{cases}$$

Differentiating $\phi(x)$ and applying (2.1) and (3.3), we obtain

$$\begin{aligned} 0 \sim \phi_\alpha &= \sum_{i,j} \frac{\partial \sigma_{l+1}}{\partial w_{ij}} w_{ij\alpha} = \sum_i \sigma_l(\lambda|i) w_{ii\alpha} \\ &= \sum_{i \in G} \sigma_l(\lambda|i) w_{ii\alpha} + \sum_{i \in B} \sigma_l(\lambda|i) w_{ii\alpha} \\ &\sim \sigma_l(G) \sum_{i \in B} w_{ii\alpha}, \end{aligned}$$

which implies

$$(3.5) \quad \sum_{i \in B} w_{ii\alpha} \sim 0.$$

Combining (3.3)-(3.5) with Lemma 2.1, for $\alpha = 1, 2, \dots, n$, we calculate the second derivatives of $\phi(x)$,

$$\begin{aligned} \phi_{\alpha\alpha} &= \sum_{i,j} \frac{\partial \sigma_{l+1}}{\partial w_{ij}} w_{ij\alpha\alpha} + \sum_{i,j,p,q} \frac{\partial^2 \sigma_{l+1}}{\partial w_{ij} \partial w_{pq}} w_{ij\alpha} w_{pq\alpha} \\ &= \sum_i \sigma_l(\lambda|i) w_{ii\alpha\alpha} + \sum_{i \neq j} \sigma_{l-1}(\lambda|ij) w_{ii\alpha} w_{jj\alpha} - \sum_{i \neq j} \sigma_{l-1}(\lambda|ij) w_{ij\alpha}^2 \\ &\sim \sigma_l(G) \sum_{i \in B} w_{ii\alpha\alpha} + \left(\sum_{\substack{i \in G \\ j \in B}} + \sum_{\substack{i \in B \\ j \in G}} + \sum_{\substack{i,j \in G \\ i \neq j}} + \sum_{\substack{i,j \in B \\ i \neq j}} \right) \sigma_{l-1}(\lambda|ij) w_{ii\alpha} w_{jj\alpha} \\ &\quad - \left(\sum_{\substack{i \in G \\ j \in B}} + \sum_{\substack{i \in B \\ j \in G}} + \sum_{\substack{i,j \in G \\ i \neq j}} + \sum_{\substack{i,j \in B \\ i \neq j}} \right) \sigma_{l-1}(\lambda|ij) w_{ij\alpha}^2 \\ &\sim \sigma_l(G) \sum_{i \in B} w_{ii\alpha\alpha} + \sum_{i \in G} \sigma_{l-1}(G|i) w_{ii\alpha} \sum_{j \in B} w_{jj\alpha} + \sum_{i \in B} w_{ii\alpha} \sum_{j \in G} \sigma_{l-1}(G|j) w_{jj\alpha} \\ &\quad + \sigma_{l-1}(G) \sum_{\substack{i,j \in B \\ i \neq j}} w_{ii\alpha} w_{jj\alpha} - 2 \sum_{\substack{i \in G \\ j \in B}} \sigma_{l-1}(G|i) w_{ij\alpha}^2 - \sigma_{l-1}(G) \sum_{\substack{i,j \in B \\ i \neq j}} w_{ij\alpha}^2 \\ &\sim \sigma_l(G) \sum_{i \in B} w_{ii\alpha\alpha} + \sigma_{l-1}(G) \sum_{\substack{i,j \in B \\ i \neq j}} w_{ii\alpha} w_{jj\alpha} - 2 \sum_{\substack{i \in G \\ j \in B}} \sigma_{l-1}(G|i) w_{ij\alpha}^2 - \sigma_{l-1}(G) \sum_{\substack{i,j \in B \\ i \neq j}} w_{ij\alpha}^2. \end{aligned}$$

From condition (3.5), we have $-w_{ii\alpha} \sim \sum_{\substack{j \in B \\ j \neq i}} w_{jj\alpha}$, thus

$$\sigma_{l-1}(G) \sum_{\substack{i,j \in B \\ i \neq j}} w_{ii\alpha} w_{jj\alpha} \sim -\sigma_{l-1}(G) \sum_{i \in B} w_{ii\alpha}^2.$$

By the commutation formula for covariant derivatives on the sphere, we have $w_{ii\alpha\alpha} = w_{\alpha\alpha ii} + w_{ii} - w_{\alpha\alpha}$, which leads to

$$\phi_{\alpha\alpha} \sim \sigma_l(G) \sum_{i \in B} (w_{\alpha\alpha ii} - w_{\alpha\alpha}) - \sigma_{l-1}(G) \sum_{i,j \in B} w_{ij\alpha}^2 - 2 \sum_{\substack{i \in G \\ j \in B}} \sigma_{l-1}(G|i) w_{ij\alpha}^2.$$

It follows from Lemma 2.2 that

$$\begin{aligned} \sum_{\alpha} F^{\alpha\alpha} \phi_{\alpha\alpha} &\sim \sigma_l(G) \sum_{\alpha} \sum_{i \in B} F^{\alpha\alpha} w_{\alpha\alpha ii} - (n-l) \sigma_l(G) \sum_{\alpha \in G} F^{\alpha\alpha} w_{\alpha\alpha} \\ (3.6) \quad &\quad - \sigma_{l-1}(G) \sum_{\alpha} \sum_{i,j \in B} F^{\alpha\alpha} w_{ij\alpha}^2 - 2 \sum_{\alpha} \sum_{\substack{i \in G \\ j \in B}} \sigma_{l-1}(G|i) F^{\alpha\alpha} w_{ij\alpha}^2. \end{aligned}$$

To handle the fourth-order derivative terms in (3.6), we differentiate both sides of the equation $F(W) = \Theta(x)$ with respect to x_i , yielding

$$(3.7) \quad \sum_{\alpha} F^{\alpha\alpha} w_{\alpha\alpha i} = \Theta_i.$$

Taking a second derivative, we arrive at

$$(3.8) \quad \sum_{\alpha, \beta, \gamma, \delta} F^{\alpha\beta, \gamma\delta} w_{\alpha\beta i} w_{\gamma\delta i} + \sum_{\alpha} F^{\alpha\alpha} w_{\alpha\alpha ii} = \Theta_{ii}.$$

Here, we have used the fact that $F^{\alpha\beta}$ is diagonal at the point x . Then, by Lemma 2.2, (3.8) implies

$$(3.9) \quad \sum_{\alpha} F^{\alpha\alpha} w_{\alpha\alpha ii} = - \sum_{\alpha} F^{\alpha\alpha, \alpha\alpha} w_{\alpha\alpha i}^2 - \sum_{\alpha \neq \beta} F^{\alpha\beta, \beta\alpha} w_{\alpha\beta i}^2 + \Theta_{ii}.$$

Substituting (3.9) into (3.6), we conclude that

$$(3.10) \quad \begin{aligned} \sum_{\alpha} F^{\alpha\alpha} \phi_{\alpha\alpha} &\sim - \sigma_l(G) \sum_{\alpha} \sum_{i \in B} F^{\alpha\alpha, \alpha\alpha} w_{\alpha\alpha i}^2 - \sigma_l(G) \sum_{\alpha \neq \beta} \sum_{i \in B} F^{\alpha\beta, \beta\alpha} w_{\alpha\beta i}^2 \\ &\quad - 2 \sum_{\alpha} \sum_{\substack{i \in G \\ j \in B}} \sigma_{l-1}(G|i) F^{\alpha\alpha} w_{ij\alpha}^2 - \sigma_{l-1}(G) \sum_{\alpha} \sum_{i, j \in B} F^{\alpha\alpha} w_{ij\alpha}^2 \\ &\quad - (n-l) \sigma_l(G) \sum_{\alpha \in G} F^{\alpha\alpha} w_{\alpha\alpha} + \sigma_l(G) \sum_{i \in B} \Theta_{ii}. \end{aligned}$$

Next, we need to carefully analyze the third-order derivative terms with respect to w on the right-hand side of (3.10), specifically the first four terms, which we denote as ①–④. It is clear that ④ is non-positive. For the term ②, from Lemma 2.2, we deduce that

$$\begin{aligned} \textcircled{2} &= - \sigma_l(G) \sum_{\alpha \neq \beta} \sum_{i \in B} F^{\alpha\beta, \beta\alpha} w_{\alpha\beta i}^2 \\ &= - \sigma_l(G) \left(\sum_{\substack{\alpha \in G \\ \beta \in B}} + \sum_{\substack{\alpha \in B \\ \beta \in G}} + \sum_{\substack{\alpha, \beta \in G \\ \alpha \neq \beta}} + \sum_{\substack{\alpha, \beta \in B \\ \alpha \neq \beta}} \right) \sum_{i \in B} F^{\alpha\beta, \beta\alpha} w_{\alpha\beta i}^2 \\ &\sim 2 \sigma_l(G) \sum_{\substack{\alpha \in G \\ \beta \in B}} \sum_{i \in B} \frac{\lambda_{\alpha}}{1 + \lambda_{\alpha}^2} w_{\alpha\beta i}^2 + \sigma_l(G) \sum_{\substack{\alpha, \beta \in G \\ \alpha \neq \beta}} \sum_{i \in B} \frac{\lambda_{\alpha} + \lambda_{\beta}}{(1 + \lambda_{\alpha}^2)(1 + \lambda_{\beta}^2)} w_{\alpha\beta i}^2 \\ &\triangleq \textcircled{2}_1 + \textcircled{2}_2. \end{aligned}$$

Further, we analyze ③ and obtain

$$\textcircled{3} = - 2 \sum_{\alpha} \sum_{\substack{i \in G \\ j \in B}} \sigma_{l-1}(G|i) F^{\alpha\alpha} w_{ij\alpha}^2$$

$$\begin{aligned}
& \sim -2 \sum_{\alpha \in G} \sum_{\substack{i \in G \\ j \in B}} \sigma_{l-1}(G|i) \frac{1}{1 + \lambda_\alpha^2} w_{ij\alpha}^2 - 2 \sum_{\alpha \in B} \sum_{\substack{i \in G \\ j \in B}} \sigma_{l-1}(G|i) w_{ij\alpha}^2 \\
& = -2 \sum_{i \in B} \sum_{\substack{\alpha, \beta \in G \\ \alpha \neq \beta}} \sigma_{l-1}(G|\beta) \frac{1}{1 + \lambda_\alpha^2} w_{\alpha\beta i}^2 - 2 \sum_{i \in B} \sum_{\alpha \in G} \sigma_{l-1}(G|\alpha) \frac{1}{1 + \lambda_\alpha^2} w_{\alpha\alpha i}^2 \\
& \quad - 2 \sum_{i \in B} \sum_{\substack{\alpha \in G \\ \beta \in B}} \sigma_{l-1}(G|\alpha) w_{\alpha\beta i}^2 \\
& \triangleq \textcircled{3}_1 + \textcircled{3}_2 + \textcircled{3}_3.
\end{aligned}$$

Combining $\textcircled{2}_1$ and $\textcircled{3}_3$, we get

$$\begin{aligned}
(3.11) \quad \textcircled{2}_1 + \textcircled{3}_3 &= 2 \sum_{i \in B} \sum_{\substack{\alpha \in G \\ \beta \in B}} \left(\sigma_l(G) \frac{\lambda_\alpha}{1 + \lambda_\alpha^2} - \sigma_{l-1}(G|\alpha) \right) w_{\alpha\beta i}^2 \\
&= -2 \sigma_l(G) \sum_{i \in B} \sum_{\substack{\alpha \in G \\ \beta \in B}} \frac{1}{\lambda_\alpha(1 + \lambda_\alpha^2)} w_{\alpha\beta i}^2 \\
&\leq 0.
\end{aligned}$$

For the terms $\textcircled{2}_2$ and $\textcircled{3}_1$, the computation gives

$$\begin{aligned}
(3.12) \quad \textcircled{2}_2 + \textcircled{3}_1 &= \sigma_l(G) \sum_{i \in B} \sum_{\substack{\alpha, \beta \in G \\ \alpha \neq \beta}} \frac{\lambda_\alpha + \lambda_\beta}{(1 + \lambda_\alpha^2)(1 + \lambda_\beta^2)} w_{\alpha\beta i}^2 - 2 \sum_{i \in B} \sum_{\substack{\alpha, \beta \in G \\ \alpha \neq \beta}} \sigma_{l-1}(G|\beta) \frac{1}{1 + \lambda_\alpha^2} w_{\alpha\beta i}^2 \\
&= \sigma_l(G) \sum_{i \in B} \sum_{\substack{\alpha, \beta \in G \\ \alpha \neq \beta}} \frac{\lambda_\alpha + \lambda_\beta}{(1 + \lambda_\alpha^2)(1 + \lambda_\beta^2)} w_{\alpha\beta i}^2 - \sum_{i \in B} \sum_{\substack{\alpha, \beta \in G \\ \alpha \neq \beta}} \sigma_{l-1}(G|\beta) \frac{1}{1 + \lambda_\alpha^2} w_{\alpha\beta i}^2 \\
&\quad - \sum_{i \in B} \sum_{\substack{\alpha, \beta \in G \\ \alpha \neq \beta}} \sigma_{l-1}(G|\alpha) \frac{1}{1 + \lambda_\beta^2} w_{\alpha\beta i}^2 \\
&= -\sigma_l(G) \sum_{i \in B} \sum_{\substack{\alpha, \beta \in G \\ \alpha \neq \beta}} \frac{\lambda_\alpha + \lambda_\beta}{\lambda_\alpha \lambda_\beta (1 + \lambda_\alpha^2)(1 + \lambda_\beta^2)} w_{\alpha\beta i}^2 \\
&\leq 0.
\end{aligned}$$

Noting that

$$\begin{aligned}
(3.13) \quad \textcircled{1} + \textcircled{3}_2 &\sim \sum_{i \in B} \sum_{\alpha \in G} \left(\sigma_l(G) \frac{2\lambda_\alpha}{(1 + \lambda_\alpha^2)^2} - \sigma_{l-1}(G|\alpha) \frac{2}{1 + \lambda_\alpha^2} \right) w_{\alpha\alpha i}^2 \\
&= -\sigma_l(G) \sum_{i \in B} \sum_{\alpha \in G} \frac{2}{\lambda_\alpha(1 + \lambda_\alpha^2)^2} w_{\alpha\alpha i}^2.
\end{aligned}$$

Thus, we can conclude from combining (3.11), (3.12) and (3.13) that

$$\begin{aligned} \sum_{\alpha} F^{\alpha\alpha} \phi_{\alpha\alpha} &\lesssim -\sigma_l(G) \sum_{i \in B} \sum_{\alpha \in G} \frac{2}{\lambda_{\alpha}(1 + \lambda_{\alpha}^2)^2} w_{\alpha\alpha i}^2 - (n-l)\sigma_l(G) \sum_{\alpha \in G} F^{\alpha\alpha} w_{\alpha\alpha} \\ &\quad + \sigma_l(G) \sum_{i \in B} \Theta_{ii}. \end{aligned}$$

Equivalently,

$$(3.14) \quad \frac{1}{\sigma_l(G)} \sum_{\alpha} F^{\alpha\alpha} \phi_{\alpha\alpha} \lesssim - \sum_{i \in B} \sum_{\alpha \in G} \frac{2}{\lambda_{\alpha}(1 + \lambda_{\alpha}^2)^2} w_{\alpha\alpha i}^2 - (n-l) \sum_{\alpha \in G} F^{\alpha\alpha} w_{\alpha\alpha} + \sum_{i \in B} \Theta_{ii}.$$

We now aim to determine the maximum of the first term on the right-hand side of (3.14), under the constraint given by (3.7), using the Lagrange multiplier technique. For simplicity of notations, fix $i \in B = \{l+1, l+2, \dots, n\}$, and for any $\alpha \in G = \{1, 2, \dots, l\}$, let $X_{\alpha} = w_{\alpha\alpha i}$. We consider the following objective function

$$(3.15) \quad f(X_1, X_2, \dots, X_l) = - \sum_{\alpha \in G} \frac{2}{\lambda_{\alpha}(1 + \lambda_{\alpha}^2)^2} X_{\alpha}^2,$$

under the constraint

$$\sum_{\alpha \in G} \frac{1}{1 + \lambda_{\alpha}^2} X_{\alpha} \sim \Theta_i,$$

and determine its maximum. Let

$$(3.16) \quad g(X_1, X_2, \dots, X_l) = \sum_{\alpha \in G} \frac{1}{1 + \lambda_{\alpha}^2} X_{\alpha} - \Theta_i \sim 0.$$

We define the Lagrangian L by

$$\begin{aligned} L(X_1, X_2, \dots, X_l, s) &= f(X_1, X_2, \dots, X_l) + sg(X_1, X_2, \dots, X_l) \\ &\sim - \sum_{\alpha \in G} \frac{2}{\lambda_{\alpha}(1 + \lambda_{\alpha}^2)^2} X_{\alpha}^2 + \sum_{\alpha \in G} \frac{s}{1 + \lambda_{\alpha}^2} X_{\alpha} - s\Theta_i, \end{aligned}$$

where $s \in \mathbb{R}$. Taking partial derivatives for L , we have, for $\alpha = 1, 2, \dots, l$,

$$\frac{\partial L}{\partial X_{\alpha}} = - \frac{4}{\lambda_{\alpha}(1 + \lambda_{\alpha}^2)^2} X_{\alpha} + \frac{s}{1 + \lambda_{\alpha}^2},$$

and

$$\frac{\partial L}{\partial s} = \sum_{\alpha \in G} \frac{1}{1 + \lambda_{\alpha}^2} X_{\alpha} - \Theta_i.$$

Let

$$\frac{\partial L}{\partial X_{\alpha}} = 0, \quad \alpha = 1, 2, \dots, l, \quad \text{and} \quad \frac{\partial L}{\partial s} \sim 0,$$

we get

$$X_{\alpha} \sim \lambda_{\alpha}(1 + \lambda_{\alpha}^2) \left(\sum_{\beta \in G} \lambda_{\beta} \right)^{-1} \Theta_i.$$

This implies that the point

$$(3.17) \quad \left(\sum_{\beta \in G} \lambda_\beta \right)^{-1} \Theta_i \left(\lambda_1(1 + \lambda_1^2), \lambda_2(1 + \lambda_2^2), \dots, \lambda_l(1 + \lambda_l^2) \right)$$

is a stationary point of the function $f(X_1, X_2, \dots, X_l)$ under the constraint (3.16).

To show that point (3.17) maximizes $f(X_1, X_2, \dots, X_l)$ (see (3.15)) subject to the constraint (3.16), we reformulate it as an unconstrained optimization problem. From (3.16), we derive

$$(3.18) \quad \frac{1}{(1 + \lambda_1^2)^2} X_1^2 \sim \Theta_i^2 - 2\Theta_i \sum_{\alpha=2}^l \frac{1}{1 + \lambda_\alpha^2} X_\alpha + \left(\sum_{\alpha=2}^l \frac{1}{1 + \lambda_\alpha^2} X_\alpha \right)^2.$$

Substituting (3.18) into (3.15) gives

$$\begin{aligned} \mathcal{F}(X_2, X_3, \dots, X_l) \sim & -2 \sum_{\alpha=2}^l \frac{\lambda_1 + \lambda_\alpha}{\lambda_1 \lambda_\alpha (1 + \lambda_\alpha^2)^2} X_\alpha^2 - \frac{2}{\lambda_1} \sum_{\substack{\alpha \neq \beta \\ \alpha, \beta = 2, 3, \dots, l}} \frac{1}{(1 + \lambda_\alpha^2)(1 + \lambda_\beta^2)} X_\alpha X_\beta \\ & + \Theta_i \sum_{\alpha=2}^l \frac{4}{\lambda_1(1 + \lambda_\alpha^2)} X_\alpha - \frac{2}{\lambda_1} \Theta_i^2. \end{aligned}$$

The Hessian matrix of $\mathcal{F}(X_2, X_3, \dots, X_l)$ is thus given by

$$M = \begin{pmatrix} -\frac{4(\lambda_1 + \lambda_2)}{\lambda_1 \lambda_2 (1 + \lambda_2^2)^2} & -\frac{2}{\lambda_1(1 + \lambda_2^2)(1 + \lambda_3^2)} & -\frac{2}{\lambda_1(1 + \lambda_2^2)(1 + \lambda_4^2)} & \cdots & -\frac{2}{\lambda_1(1 + \lambda_2^2)(1 + \lambda_l^2)} \\ -\frac{2}{\lambda_1(1 + \lambda_2^2)(1 + \lambda_3^2)} & -\frac{4(\lambda_1 + \lambda_3)}{\lambda_1 \lambda_3 (1 + \lambda_3^2)^2} & -\frac{2}{\lambda_1(1 + \lambda_3^2)(1 + \lambda_4^2)} & \cdots & -\frac{2}{\lambda_1(1 + \lambda_3^2)(1 + \lambda_l^2)} \\ -\frac{2}{\lambda_1(1 + \lambda_2^2)(1 + \lambda_4^2)} & -\frac{2}{\lambda_1(1 + \lambda_3^2)(1 + \lambda_4^2)} & -\frac{4(\lambda_1 + \lambda_4)}{\lambda_1 \lambda_4 (1 + \lambda_4^2)^2} & \cdots & -\frac{2}{\lambda_1(1 + \lambda_4^2)(1 + \lambda_l^2)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -\frac{2}{\lambda_1(1 + \lambda_2^2)(1 + \lambda_l^2)} & -\frac{2}{\lambda_1(1 + \lambda_3^2)(1 + \lambda_l^2)} & -\frac{2}{\lambda_1(1 + \lambda_4^2)(1 + \lambda_l^2)} & \cdots & -\frac{4(\lambda_1 + \lambda_l)}{\lambda_1 \lambda_l (1 + \lambda_l^2)^2} \end{pmatrix}_{l-1 \times l-1}.$$

It now suffices to verify that M is negative definite. Equivalently, we may show that the matrix

$$M' = \begin{pmatrix} \frac{2(\lambda_1 + \lambda_2)}{\lambda_2(1 + \lambda_2^2)^2} & \frac{1}{(1 + \lambda_2^2)(1 + \lambda_3^2)} & \frac{1}{(1 + \lambda_2^2)(1 + \lambda_4^2)} & \cdots & \frac{1}{(1 + \lambda_2^2)(1 + \lambda_l^2)} \\ \frac{1}{(1 + \lambda_2^2)(1 + \lambda_3^2)} & \frac{2(\lambda_1 + \lambda_3)}{\lambda_3(1 + \lambda_3^2)^2} & \frac{1}{(1 + \lambda_3^2)(1 + \lambda_4^2)} & \cdots & \frac{1}{(1 + \lambda_3^2)(1 + \lambda_l^2)} \\ \frac{1}{(1 + \lambda_2^2)(1 + \lambda_4^2)} & \frac{1}{(1 + \lambda_3^2)(1 + \lambda_4^2)} & \frac{2(\lambda_1 + \lambda_4)}{\lambda_4(1 + \lambda_4^2)^2} & \cdots & \frac{1}{(1 + \lambda_4^2)(1 + \lambda_l^2)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{1}{(1 + \lambda_2^2)(1 + \lambda_l^2)} & \frac{1}{(1 + \lambda_3^2)(1 + \lambda_l^2)} & \frac{1}{(1 + \lambda_4^2)(1 + \lambda_l^2)} & \cdots & \frac{2(\lambda_1 + \lambda_l)}{\lambda_l(1 + \lambda_l^2)^2} \end{pmatrix}_{l-1 \times l-1}$$

is positive definite. This reduces to checking that all $(m-1)$ -th order leading principal minors $\det(M'_{m-1})$ of M' are positive for $m = 2, 3, \dots, l$. Obviously,

$$M'_{m-1} = A + vv^\top,$$

where

$$A = \text{diag} \left(\frac{2\lambda_1 + \lambda_2}{\lambda_2(1 + \lambda_2^2)^2}, \frac{2\lambda_1 + \lambda_3}{\lambda_3(1 + \lambda_3^2)^2}, \dots, \frac{2\lambda_1 + \lambda_m}{\lambda_m(1 + \lambda_m^2)^2} \right)$$

and

$$v^\top = \left(\frac{1}{1 + \lambda_2^2}, \frac{1}{1 + \lambda_3^2}, \dots, \frac{1}{1 + \lambda_m^2} \right).$$

It follows immediately from Lemma 2.3 that

$$\det(M'_{m-1}) = \left(\prod_{i=2}^m \frac{2\lambda_1 + \lambda_i}{\lambda_i(1 + \lambda_i^2)^2} \right) \left(1 + \sum_{i=2}^m \frac{\lambda_i}{2\lambda_1 + \lambda_i} \right) > 0$$

for all $m = 2, 3, \dots, l$.

With this, we have shown that

$$(3.19) \quad f(X_1, X_2, \dots, X_l) \lesssim -2 \left(\sum_{\alpha \in G} \lambda_\alpha \right)^{-1} \Theta_i^2$$

subject to (3.16). Putting (3.19) into (3.14), we have

$$(3.20) \quad \frac{1}{\sigma_l(G)} \sum_{\alpha} F^{\alpha\alpha} \phi_{\alpha\alpha} \lesssim \sum_{i \in B} \left(-2 \left(\sum_{\alpha \in G} \lambda_\alpha \right)^{-1} \Theta_i^2 + \Theta_{ii} - \sum_{\alpha \in G} F^{\alpha\alpha} w_{\alpha\alpha} \right).$$

We denote $\theta_\alpha = \arctan \lambda_\alpha$ for $\alpha = 1, 2, \dots, l$, $\lambda_\alpha = \tan \theta_\alpha$ and $\Theta \sim \sum_{\alpha=1}^l \theta_\alpha$. The last term on the right-hand side of (3.20) can evidently be expressed as

$$\sum_{\alpha \in G} F^{\alpha\alpha} w_{\alpha\alpha} = \sum_{\alpha=1}^l \frac{\tan \theta_\alpha}{1 + \tan^2 \theta_\alpha} = \frac{1}{2} \sum_{\alpha=1}^l \sin 2\theta_\alpha.$$

Since $\Theta \in (0, \pi/2)$ and $\theta_\alpha > 0$ for any $\alpha = 1, 2, \dots, l$, by using induction, we arrive at

$$\begin{aligned}
\frac{1}{2} \sin \left(2 \sum_{\alpha=1}^l \theta_\alpha \right) &= \frac{1}{2} \sin \left(2\theta_1 + 2 \sum_{\alpha=2}^l \theta_\alpha \right) \\
&= \frac{1}{2} \sin 2\theta_1 \cos \left(2 \sum_{\alpha=2}^l \theta_\alpha \right) + \frac{1}{2} \cos 2\theta_1 \sin \left(2 \sum_{\alpha=2}^l \theta_\alpha \right) \\
&\leq \frac{1}{2} \sin 2\theta_1 + \frac{1}{2} \sin \left(2\theta_2 + 2 \sum_{\alpha=3}^l \theta_\alpha \right) \\
&\leq \dots \\
&\leq \frac{1}{2} \sum_{\alpha=1}^l \sin 2\theta_\alpha.
\end{aligned}$$

Given $\frac{1}{2} \sin(2 \sum_{\alpha=1}^l \theta_\alpha) \sim \frac{\tan \Theta}{\sec^2 \Theta}$, we can deduce that

$$(3.21) \quad \frac{\tan \Theta}{\sec^2 \Theta} \lesssim \sum_{\alpha \in G} F^{\alpha\alpha} w_{\alpha\alpha}.$$

Substituting (3.21) into (3.20), we obtain

$$(3.22) \quad \frac{1}{\sigma_l(G)} \sum_{\alpha} F^{\alpha\alpha} \phi_{\alpha\alpha} \lesssim \sum_{i \in B} \left(-2 \left(\sum_{\alpha \in G} \lambda_\alpha \right)^{-1} \Theta_i^2 + \Theta_{ii} - \frac{\tan \Theta}{\sec^2 \Theta} \right).$$

To obtain (3.1), we state the following claim and prove it.

Claim: If $\Theta \in (0, \pi/2)$, then

$$(3.23) \quad - \left(\sum_{\alpha \in G} \lambda_\alpha \right)^{-1} \lesssim -\frac{1}{\tan \Theta}, \quad G = \{1, 2, \dots, l\}.$$

Clearly, (3.23) is equivalent to

$$\sum_{\alpha \in G} \tan \theta_\alpha \lesssim \tan \Theta,$$

so we only need to prove this. Since $\Theta \in (0, \pi/2)$, for any choice of θ_β and θ_γ , where $\beta, \gamma = 1, 2, \dots, l$ and $\beta \neq \gamma$, we get $\tan \theta_\beta > 0$, $\tan \theta_\gamma > 0$ and $\tan(\theta_\beta + \theta_\gamma) > 0$. Therefore, $\tan(\theta_\beta + \theta_\gamma) = \frac{\tan \theta_\beta + \tan \theta_\gamma}{1 - \tan \theta_\beta \tan \theta_\gamma}$ implies that

$$\tan \theta_\beta + \tan \theta_\gamma \leq \tan(\theta_\beta + \theta_\gamma).$$

Following this, we can obtain

$$\sum_{\alpha \in G} \tan \theta_\alpha \leq \tan \left(\sum_{\alpha \in G} \theta_\alpha \right) \sim \tan \Theta.$$

With this, we have completed the proof of **Claim**.

From (3.22) and (3.23),

$$(3.24) \quad \frac{1}{\sigma_l(G)} \sum_{\alpha} F^{\alpha\alpha} \phi_{\alpha\alpha} \lesssim \sum_{i \in B} \left(-\frac{2}{\tan \Theta} \Theta_i^2 + \Theta_{ii} - \frac{\tan \Theta}{\sec^2 \Theta} \right).$$

Because of

$$\left(\frac{1}{\tan \Theta} \right)_{ii} = -\frac{\sec^2 \Theta}{\tan^2 \Theta} \left(-\frac{2}{\tan \Theta} \Theta_i^2 + \Theta_{ii} \right),$$

(3.24) yields

$$\frac{1}{\sigma_l(G)} \sum_{\alpha} F^{\alpha\alpha} \phi_{\alpha\alpha} \lesssim -\frac{\tan^2 \Theta}{\sec^2 \Theta} \sum_{i \in B} \left(\left(\frac{1}{\tan \Theta} \right)_{ii} + \frac{1}{\tan \Theta} \right).$$

Under our assumption that $((1/\tan \Theta)_{ij} + (1/\tan \Theta)\delta_{ij})$ is semi-positive definite matrix, it follows that

$$\sum_{\alpha} F^{\alpha\alpha} \phi_{\alpha\alpha} \lesssim 0.$$

□

Let us now give the proof of Theorem 1.1.

Proof of Theorem 1.1. From the differential inequality (3.1) in Lemma 3.1 and the strong maximum principle, we conclude that $\phi = \sigma_{l+1}(W) \equiv 0$ on $\mathbb{S}^n$, where l denotes the minimal rank of the spherical Hessian matrix W . This means that W has constant rank on $\mathbb{S}^n$.

Suppose without loss of generality that $\text{rank}(W) = l < n$, so that

$$\sigma_l(W) > 0 \quad \text{and} \quad \sigma_{l+1}(W) = 0 \quad \text{on } \mathbb{S}^n.$$

However, applying the Minkowski integral formula (see [18])

$$(3.25) \quad (n-l) \int_{\mathbb{S}^n} u \sigma_l(W) = (l+1) \int_{\mathbb{S}^n} \sigma_{l+1}(W),$$

we arrive at a contradiction, since the left-hand side of (3.25) is strictly positive while the right-hand side is vanishing. Therefore, W must be of full rank on $\mathbb{S}^n$, and hence is positive definite. □

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